Key Insights and Final Recommendations for Tobe Energy

The Ronnie K. Irani Center for the Creation of Economic Wealth

Energy Fall 2025

Consulting Associates

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Executive Summary

This semester, the Ronnie K. Irani Center for the Creation of Economic Wealth has provided five consulting interns with the opportunity to work with Tobe, an Oklahoma-based startup focused on making green hydrogen feasible and affordable. Tobe manufactures electrolyzer machines in three sizes (25kg/day, 250kg/day, 2500kg/day), which operate at a 94.7% efficiency, and produce hydrogen at ~$446 per kWh. Tobe's electrolyzer requires an input of water and electricity, and outputs H_2. Through the team's research, including extensive market analysis and 7 interviews with industry experts and professionals, the Energy Team has analyzed Tobe's market viability, business model, and growth needs. The team examined the demand for hydrogen as a service (HaaS) versus electrolyzer sales. The team determined who Tobe's main competitors in the hydrogen space could be, and how Tobe can position themselves within the market to provide services that no competitor is currently offering.

Tobe's ideal customer is hydrogen users that are 150+ miles away from their hydrogen source because cost is the highest contributing factor for companies making hydrogen decisions.[1]

- Location: Transportation costs are the highest cost factor outside of 150 miles. Tobe should concentrate on securing companies that are 150+ miles from their hydrogen source to stay cost competitive.[2]
- Green Commitments: Tobe should also pursue companies with green commitments regardless of location because cost is not the priority.[3]

Tobe's $9.81B serviceable market and 8 diversified segments, making up ~11% of the total market and over 10 million kg/year of hydrogen, provide a focused pathway to scale market share.

The 8 segments are heavy-duty fueling, urban fleet fueling, light & medium-duty commercial fueling, industrial decarbonization, chemical production & synthetic fuel, backup power & microgrids, energy storage, and already green customers. These segments are defined by qualities like hydrogen volume requirements and established infrastructure to establish use cases.

Tobe will benefit from 2 business models, Hydrogen-as-a-Service (HaaS) and Electrolyzer Sales, that can address segments based on industry factors.

[1] Interview, Dipankhar Ghosh, Appendix G
[2] Interview, Steve Syzmanski, Appendix E
[3] Interview, Ashish Bhakta, Appendix C

- Heavy-duty fueling, industrial decarbonization, chemical production & synthetic fuels: these companies are more likely to buy electrolyzers directly because they use large amounts of hydrogen and have pre-existing infrastructure to support it.[4] [5] [6]
- Light & medium-duty commercial fueling, backup power & microgrids, & energy storage/grid balancing: these companies would more likely take the HaaS model because of low hydrogen needs, seasonal hydrogen needs, or low capital funding.[7] [8]
- Urban fleet fueling & already green companies: These industries have variable factors across companies that will lead to a mix of HaaS and electrolyzer sales.[9]

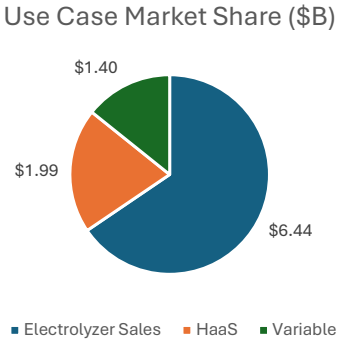

Figure 1

Tobe could attain $657.7 million in profit by 2033 by instituting a 150% profit margin on electrolyzers and a 30% profit margin on hydrogen.

- Tobe can charge a 150% profit margin on electrolyzers while staying cost competitive compared to competitors.[10] [11]
- Tobe can charge $3.51 per kg of hydrogen on the low end. If customer utility costs are low, Tobe can charge up to their current cost per kg – utilities – 15%.[12]
- Tobe can achieve a target growth rate of 50% by increasing its labor force proportionately to the number of electrolyzers needed.

[4] Linder, Truck Fleet Size Statistics, 2025
[5] Interview, Ashish Bhakta, Appendix C
[6] WHA International, Top Industrial Uses of Hydrogen, and the Need for Industrial Hydrogen Safety, 2023
[7] U.S. Department of Transportation, U.S. Automobile and Truck Fleets by Use
[8] Grand View Research, Hydrogen Energy Storage Market, 2024
[9] Interview, Ashish Bhakta, Appendix C
[10] Spindle, Bill, More Companies Moving to Adopt Decarbonization Targets
[11] Hydrogen Insight, European Companies Are Willing to Pay Premiums of More Than 50% for Green Hydrogen-Based Fuels: BNEF
[12] Interview, Amish Patel, Appendix B

Context

The hydrogen industry constituted 2.5% of the energy consumption in 2021.[13] Less than 1% of the hydrogen industry is currently considered green.[14] However, green hydrogen is projected to account for 30% of all hydrogen in 2030 and 73-100% of all hydrogen in 2050.[15] By 2050, it's predicted that hydrogen will make up 13% of the energy industry granted that the world remains under the 1.5 degrees Celsius increase limit.[16]

There are significant differences between green and grey hydrogen in the amount of carbon emissions released and the cost. Green hydrogen releases less than 1 kg of carbon emissions per kg/hydrogen while grey hydrogen releases about 12 kgs of carbon emissions.[17] Green hydrogen costs range from $4.5 to $12 per kg./hydrogen.[18] Meanwhile, grey hydrogen costs range from $1 to $3 per kg/hydrogen.[19] The cost of hydrogen is driven by economies of scale; the current environment does not have sufficient demand or supply to.[20]

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Problem

The green hydrogen market dynamic is characterized by a supply/demand dilemma.

Various common forms of hydrogen, most notably grey hydrogen, are significantly cheaper than green hydrogen.[21] Due to grey hydrogens cheaper cost, we see a supply/demand dilemma within green hydrogen industry in which there are not enough producers of green hydrogen to be cost-efficient, while at the same time there is not enough demand to drive increased production.[22]

Transportation in the hydrogen market is the largest driver of cost increases.

Though grey hydrogen is cheaper at first glance, transportation costs for users far from hydrogen supply can eradicate grey hydrogen cost advantage. The difficulty of moving hydrogen efficiently and safely is a major reason for the high transportation costs.

Pipelines tend to be the cheapest form of transportation ($0.2-0.5/kg H_2), but given their limited presence of 1,600 miles, this is only utilized by large-scale hydrogen users for in-house hydrogen transport.[23] Tube trailers transport compressed hydrogen around the 150-mile range and is

[13] IEA
[14] MIT Technology Review
[15] McKinsey
[16] Hydrogen Insight
[17] MIT Climate Portal
[18] Bloomberg NEF
[19] Bloomberg NEF
[20] Interview, Amish Patel, Appendix B
[21] Interview, Dipankar Ghosh, Appendix G
[22] Interview, Amish Patel, Appendix B
[23] Center for Sustainable Systems, University of Michigan, Hydrogen Factsheet, 2023

expensive ($0.9-1.9/kg H_2). Liquid tankers are better suited than tube trailers for transporting larger amounts of hydrogen over longer distances, but it is significantly more expensive ($2.7-3.2/kg H_2) due to energy and equipment requirements for the liquefaction process.[24] [25] This inhibitor to infrastructure alongside transportation cost barriers discourages potential users from adopting hydrogen as a fuel source.

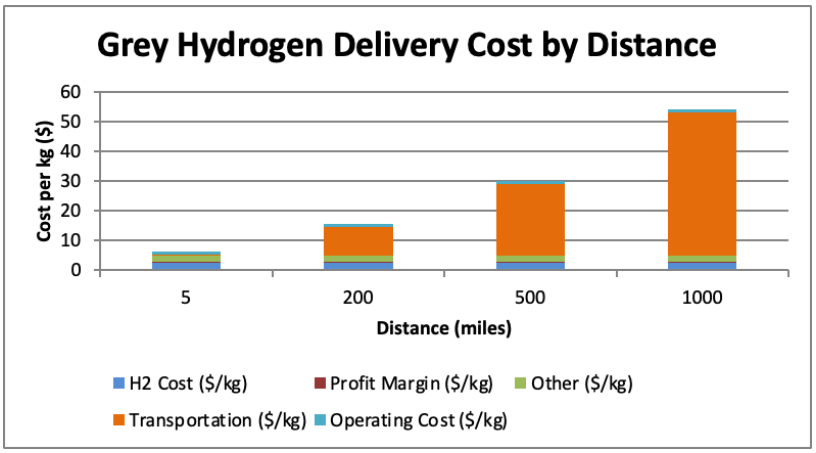

Figure 2

When the cost factors of grey hydrogen are considered per kg, the factors such as operating expenses, profit margins, cost of hydrogen, and cost of production of the hydrogen stay stable as mileage increases, as depicted in Figure 1. However, as transportation factor is broken down per mile transported, it does not stay stable but rather drives the price increase from $5 for a city located 5 miles away to over $50 for a city located 1000 miles away. We measure these cost factors that encompass the total cost for hydrogen for cities located increasingly farther away from the nearest hydrogen supply to depict the impact of transportation costs on overall grey hydrogen cost.

[24]Center for Sustainable Systems, University of Michigan, Hydrogen Factsheet, 2023
[25]Tatsutani, et. al., Techno-Economic Realities of Long-Distance Hydrogen Transport: A Cost Analysis of Importing Low-Carbon Hydrogen to Europe, 2023

Ideal Customer

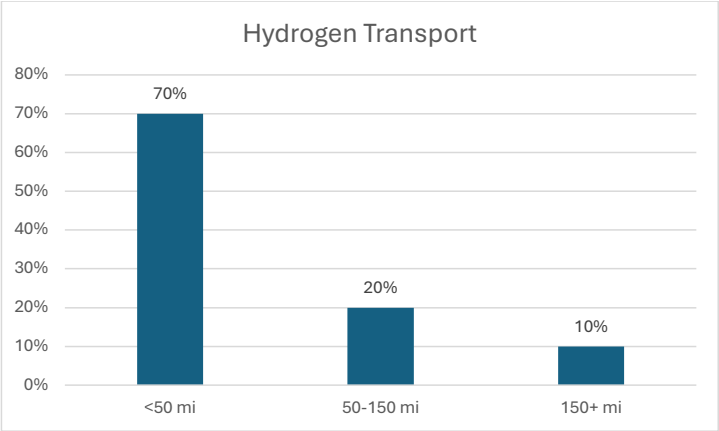

Figure 3

Tobe's ideal customers are companies located 150 miles away from hydrogen hubs and production sites because transporting hydrogen becomes costly due to convergence into liquid hydrogen.[26] Tobe offers on-site production, a cost-effective solution. Additionally, there has been a large increase in the number of companies declaring carbon-emission reduction goals with 45% of the Fortune 500 declaring emission goals and 79% of North American companies setting a net-zero target.[27] A few such companies and industries with strong green initiatives or carbon-reduction goals are also willing to pay a premium for green hydrogen, highlighting the global environment drive.[28] Customers most burdened by transportation costs and customers who have set decarbonization goals are Tobe's ideal customers who will benefit from cost-efficient and zero-carbon emission electrolyzers.

Market

Tobe's $9.81B serviceable market and 8 diversified segments, making up about 11% of the total market and over 10 million kg/year of hydrogen, provide a focused pathway to scale market share.

Market advantage exists where hydrogen demand is rising but delivery costs make grey hydrogen uneconomical, making on-site green hydrogen production the most cost-competitive

[26] Interview, Steve Syzmanski, Appendix E
[27] Spindle, Bill, More Companies Moving to Adopt Decarbonization Targets
[28] Hydrogen Insight, European Companies Are Willing to Pay Premiums of More Than 50% for Green Hydrogen-Based Fuels: BNEF

solution. Tobe's total addressable market is $90 billion, but this figure includes large portions of demand irrelevant to Tobe's commercial model such as firms who produce and use their own hydrogen on-site. Eliminating this section yields a more realistic Serviceable Available Market (SAM). From this refined market, we further narrow the SAM to U.S. demand segments where green hydrogen adoption is accelerating due to industrial decarbonization goals, transportation fuel alternatives, and newly implemented federal incentives for clean energy production. This produces a Serviceable Obtainable Market (SOM) of $9.81 billion. The SOM reflects sectors that not only show strong hydrogen growth but also are a good fit for Tobe's electrolyzers' capabilities, and hardware-plus-service business strategy.

Within this $9.81 billion SOM, the market is then segmented into eight commercially distinct clusters, each chosen because of its strategic fit with Tobe's HaaS and Electrolyzer Sales offerings. These clusters reflect differences in hydrogen usage patterns, infrastructure needs, capital budgeting behaviors, and likelihood of green hydrogen adoption. Together, they represent a combined market value of $9.81 billion served by approximately 32,025 potential customers.

The U.S. green hydrogen market is still in its early stages, but it's now characterized by accelerated growth and favorable government policies. This improves the bankability of Tobe's projects since it can now structure both its HaaS deployments and Electrolyzer Sales offerings around predictable, performance-based credits, reducing revenue uncertainty, and making project financing more feasible. Examples of these policies include hydrogen production tax credits, the most significant of which came in January 2025, when the U.S. Treasury finalized the long-awaited Section 45V production tax credit (PTC).[29] This credit provides up to $3 per kg of clean hydrogen produced, depending on verified lifecycle emissions. The final rules on emissions clarify eligibility and reporting pathways for projects that begin construction before 2033, which gives investors and developers a predictable framework for financial modeling.

Segments

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[29] Arnold & Porter (2025), 45V Clean Hydrogen Tax Credit Regulations, Environmental Edge..

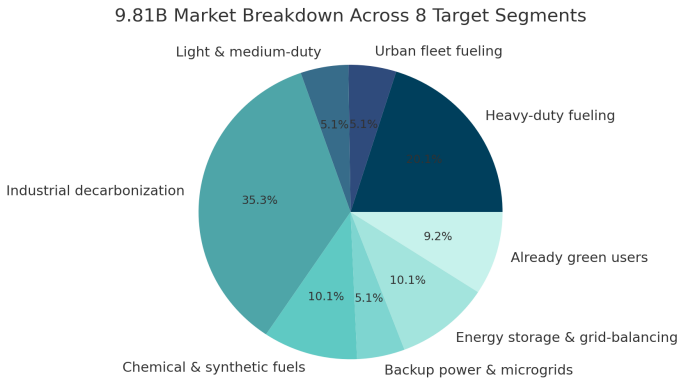

Figure 3

Our $9.81B serviceable market is distributed across eight priority customer segments. Industrial decarbonization is the largest at $3.47B, followed by heavy-duty fueling at $1.98B. Chemical production and synthetic fuels, along with energy storage and grid-balancing, each represent $0.99B opportunities. Urban fleet fueling, light and medium-duty commercial fleets, and backup power/microgrids each contribute $0.50B, reflecting strong demand from municipal fleets, last-mile delivery operators, and critical-infrastructure users. The remaining $0.90B comes from "already green" hydrogen users, where Tobe can undercut competitors by offering the most cost-effective solutions. Together, these eight segments form the most actionable portion of our market and guide our go-to-market strategy.[30] [31] [32] [33]

Segment	Average Usage (kg/day)
1. **Heavy-duty fueling**: truck stops, long-haul corridors	4,000
2. **Urban fleet fueling**: buses, municipal fleets, ports, airports	5,000
3. **Light & medium-duty commercial fueling**: last-mile delivery fleets, local distribution	200
4. **Industrial decarbonization**: electronics and semiconductor manufacturing, glass and metal processing, food and beverage hydrogenation	10,000
5. **Chemical production & synthetic fuels**: ammonia, methanol production	15,000

[30] Governor's Office of Business and Economic Development, California Launches World-Leading Hydrogen Hub, 2024
[31] Interview, Ashish Bhakta, Appendix C
[32] Bracci, et. al., Levelized Cost of Dispensed Hydrogen for Heay-Duty Vehicles, 2024
[33] Hydrogen Council, Path to Hydrogen Competitiveness: A Cost Perspective, 2020

a. Small portion of the total industry that wants carbon offsets	
6. **Backup power & microgrids**: data centers, hospital backup, remote sites	100
7. **Energy storage/grid balancing**: power-to-gas, seasonal/utility storage	1,500
8. **Already green**: Tobe can undercut competitors and gain customers already committed to green hydrogen	3,100

Table 1

Geographic Market Analysis

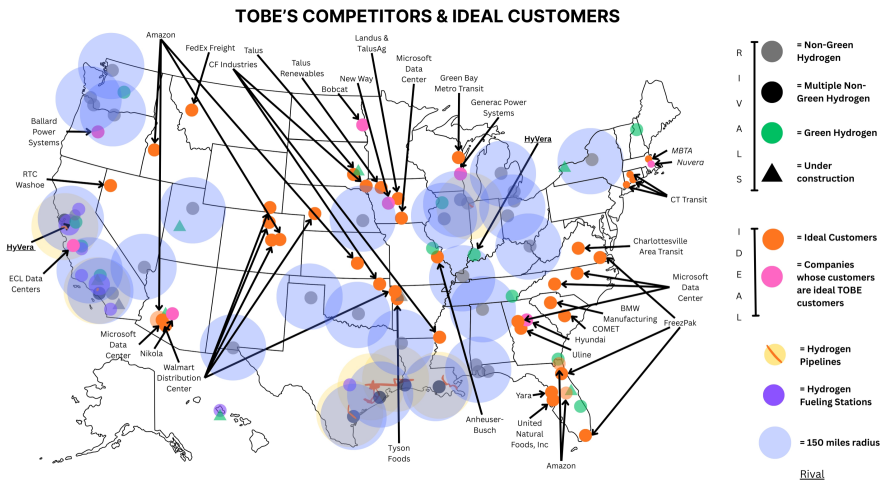

Figure 4

The map is a tool for TOBE to visualize where to expand the business. It is recommended that TOBE expands the business 150 miles away from grey hydrogen projects. The recommendation is based upon where it is most cost-effective and profitable for Tobe to operate considering locations of competitors.

The map includes locations of green and non-green hydrogen projects, potential rivals, ideal customers, companies whose customers are ideal customers for TOBE, hydrogen pipelines, green and non-green hydrogen projects under construction and hydrogen fueling stations.

The ideal customers were identified based on distance to the closest grey hydrogen project. It was calculated that customers outside the 150-mile radius benefit economically from choosing

TOBE. The map visualizes the 150-mile distance in the form of purple circles around the non-green hydrogen projects. Ideal customers are located everywhere in the country. Locations of competitors were found using multiple sources. Most data come from Clean Energy Group's Hydrogen Project Map along with a few other sources.[34] [35] [36] [37] [38] Some grey hydrogen projects were left out when creating Tobe's map as those hydrogen projects are producing for on-site use. A limited number of non-grey and non-green projects were included (pink, blue, turquoise hydrogen) but were categorized together with the grey hydrogen projects as they aren't considered green.

Many of the ideal customers use hydrogen for material handling equipment e.g. forklifts. Some ideal customers use hydrogen in their transit systems for hydrogen-powered buses. Other customers use hydrogen for trucks and ammonia production. Furthermore, companies producing products that require hydrogen were identified as their customer base are potential TOBE customers. See Appendix K for a full list of ideal customers and companies whose customers are ideal TOBE customers.

There are 1600 miles of hydrogen pipelines across the US[39]. Those are marked out by red lines with a yellow 150-mile radius. In contrast, there are 300 million miles of natural gas pipelines[40]. The hydrogen pipelines are located mostly in Texas, California, and Illinois. All hydrogen pipelines are used for feedstock and are not delivered to private customers.[41] According to industry professionals, natural gas pipelines cannot be used for hydrogen transportation without improvement.[42] Transporting hydrogen in non-hydrogen pipelines would result in the pipelines becoming brittle, and the gaskets would lose elasticity.[43]

Hydrogen hubs exist in multiple states across the country and differ from hydrogen projects. Hydrogen hubs can include multiple hydrogen projects, and some states are involved in a hydrogen hub without having a hydrogen project located in their states. Many of the hydrogen hubs across the USA received funding from the government, however, many lost their funding in October 2025.[44] While the future of the defunded hydrogen hubs is unclear, this could give

[34] Clean Energy Group. "Hydrogen Projects in the US."
[35] Manuel, Heather L. "Gevo Finalizes Renewable Hydrogen Development Services Agreement with Zero6 Energy, Selects Cummins as Technology Partner."
[36] Proctor, Darrell. "500-MW Hydrogen Power Plant Planned in Arkansas."
[37] Tavarez, Dabel. "BayoTech Celebrates Milestone Achievement: Completion of First Hydrogen Hub in Missouri."
[38] Vibbert, Meaghan. "Douglas PUD Commission Cuts Ribbon at Renewable Hydrogen Production Facility on National Hydrogen Day."
[39] US Department of Energy, "Hydrogen Pipelines."
[40] US Energy Information Administration, "Natural Gas Explained."
[41] Nygaard, Interview
[42] Nygaard, Interview
[43] Nygaard, Interview
[44] Fuel Cells Works, "DOE to Cancel Hydrogen Hub Grants, Leaked Docs Reveal."

TOBE an advantage in expanding the business. If hydrogen hubs are stalled or canceled, TOBE could expand in those areas.

Business Model

Tobe will benefit from 2 business models, Hydrogen-as-a-Service (HaaS) and Electrolyzer Sales, that can address segments based on industry factors.

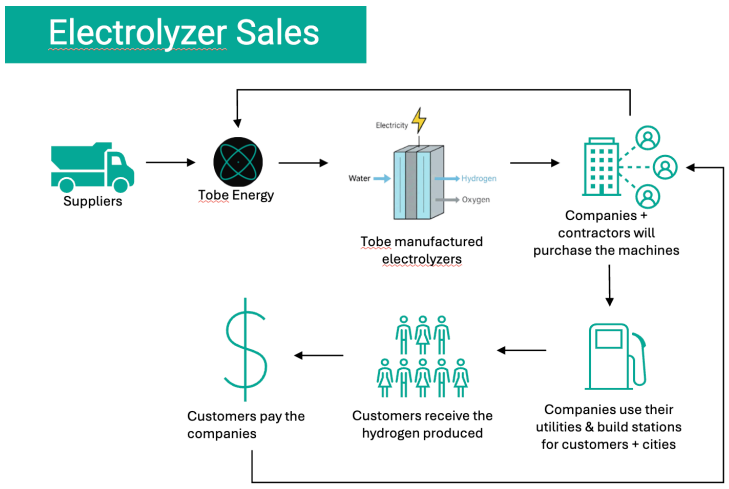

Figure 5

Electrolyzer Sales:

- Tobe manufactures and sells electrolyzers at the following costs to companies or contractors who will be using their own utilities to power the machines. In this model, Tobe will be installing the machines to the companies location.
    - 25kg electolyzer price is 43,125.00 per device
    - 250kg electolyzer price is 600,000.00 per device
    - 2500kg electolyzer price is 4,312,500.00 per device
- Customers purchase the machines directly and manage hydrogen production independently.
- Tobe profits at a 30% margin from the sale and can add optional servicing contracts for further revenue add to it such as installation, servicing, and long-term maintenance.

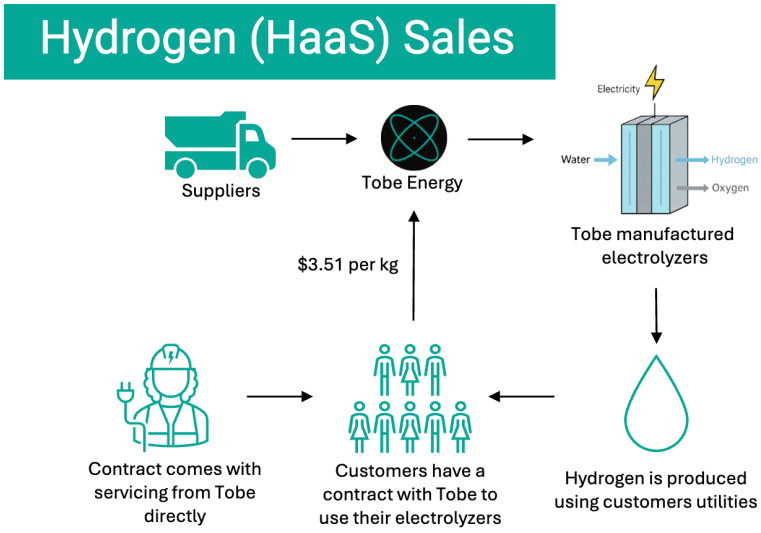

Figure 6

Hydrogen-as-a-Service (HaaS Model):

- Tobe installs electrolyzers at client sites and maintains ownership.
- Clients pay $3.51 per kg of hydrogen produced via 3-year service contracts, which include maintenance, calibration, and operational data analytics.
- This model offers recurring revenue and builds long-term relationships with industrial and transportation clients

Value Add

Tobe creates value by redesigning the full cost structure of green hydrogen rather than improving only one part of it. Their electrolyzers reduce the energy required to produce hydrogen, which directly lowers electricity expenses, but the financial benefits extend much further. Tobe's systems reduce upfront capital costs, simplify installation, and cut the ongoing maintenance burden that often makes traditional electrolyzers expensive to operate. Their modular units also reduce the need for extensive construction work and site preparation, and by producing hydrogen on-site, customers avoid the high transportation and storage costs that usually make hydrogen projects difficult to justify. The technology also reduces carbon emissions, helping customers qualify for federal clean-energy incentives, ESG benefits, and carbon credit revenue.

The value-add calculator allows companies to quantify these savings using their own project assumptions. Users can enter their expected energy consumption, electricity pricing, annual

hydrogen production, and both Tobe and competitor capital and operating costs. They can also include the value of avoided carbon emissions if they participate in carbon markets or incentive programs. Once these inputs are added, the calculator combines them to show the total economic advantage of choosing Tobe. This gives each company a clear, data-driven understanding of how much money they can save and how quickly a Tobe system would pay back in their specific situation.

This Tobe Energy value-add calculator shows how much money a customer can save by switching from a competitor's electrolyzer to Tobe's system. It compares the competitor's energy consumption (45.8 kWh/kg) to Tobe's lower usage (35.2 kWh/kg), multiplies the difference by the customer's own annual hydrogen production and local electricity price, and calculates yearly energy-cost savings. It then compares capital expenditure per kilowatt between the competitor ($4,351/kW) and Tobe ($446/kW) using the user's system size in kilowatts, revealing upfront installation savings. Operating savings come from taking the competitor's annual OPEX (e.g., $1,000,000) and applying Tobe's 45% reduction factor. The calculator also includes environmental value by multiplying carbon dioxide avoided (9,125 tons) by a carbon price ($25/ton). By changing these input values, production, electricity price, system size, CAPEX numbers, OPEX numbers, and carbon price, any customer can plug in their own data and instantly see their personalized Total Value Add, which in this example equals $1,649,280.

Financial Model

Tobe could attain $657.7 million in profit by 2033 by instituting a 150% profit margin on electrolyzers and a 30% profit margin on hydrogen.

Revenue Model

Tobe's hybrid model (HaaS + Electrolyzer-Sales) guarantees predictable revenue: HaaS captures recurring income from distance-sensitive users, while electrolyzer sales monetize large industrial users with existing infrastructure.

The HaaS model provides the foundation for recurring revenue, which stabilizes cash flow and ensures sustainable profitability as Tobe scales. Additionally, it generates steady income by producing and selling hydrogen directly to end-users at their facilities. According to Tobe's financial outlook, each partner site is modeled with an installed capacity of 2.65 MW, producing about 37,500 kgs per month, with the ability to scale up to 5.72 MW (75,000 kg/month) as demand increases. Tobe's hydrogen will sell at $3.51/kg, with production costs near $3.96/kg, yielding a 30% gross margin. By operating 330 days per year, Tobe will achieve profitability while keeping its services competitively priced.

The HaaS model ensures that Tobe can increase revenue without increasing capital risk, which is advantageous for a rollout aiming to gain traction in the early years. Market assumptions project

an initial reach of 20% of the ideal, targetable customer pool in Year 1, expanding to 60% by Year 5, supported by a 10% annual market growth rate. Customers pay through a blended service fee ($0.02 per day annually) - combining fixed equipment uptime payments and variable per-kg charges based on how much hydrogen is produced by Tobe‘s electrolyzers.

Prioritizing early deployment in mobility and distributed power segments enables Tobe to acquire early adopters as their first customers and achieve volume-driven profitability. This should give Tobe the benefit of economies of scale and lower per-kg production costs faster than its competitors. The strongest HaaS opportunities are in urban and fleet fueling, backup power, and energy storage, which together represent about $2.35 billion in accessible market value and 21,000 potential customers. These sectors have high energy needs (which translates to high need for hydrogen as a potential fuel) and including them in the rollout plan should reduce revenue volatility and simplify logistics management across multiple sites. Locking customers into long-term service contracts increases credibility and Tobe’s potential to attract investors, both of which are key to funding future expansion. Each HaaS site would operate under a 3-year contract, blending fixed lease payments with per-kg usage fees.

Clients would commit to monthly minimum volumes that ensure baseline revenue, while uptime guarantees and integrated O&M would build trust.

Haas Revenue		
H2 Price (including compression)	3.51	per kg
25kg Installation fee	1,500.00	per device per contract
250kg Installation fee	15,000.00	per device per contract
2500kg Installation fee	150,000.00	per device per contract
Device Sales Revenue		
25kg Device price	43,125.00	per device
250kg Device price	600,000.00	per device
2500kg Device price	4,312,500.00	per device
25kg Device service	862.50	per d per year
250kg Device service	12,000.00	per d per year
2500kg Device service	86,250.00	per d per year

Table 2

Structuring the model around a variety of products creates a balanced, scalable sales model that enables early liquidity and lays the foundation for long-term relationships with customers. Prioritizing high-consumption sectors ensures Tobe anchors its rollout in markets which would

deliver immediate revenue. The largest opportunities for electrolyzer sales lie in industrial decarbonization (electronics, glass, and food processing), chemical and synthetic fuel production, and light commercial fueling, totaling about $6.56 billion in reachable market value and 10,870 potential customers.

Cost Structure

For the HaaS model, the capital expenditure (CAPEX) includes electrolyzer equipment, compression and purification systems, RO water treatment, and facility installation. Breaking down OPEX categories allows Tobe to control recurring costs and maintain its 30% gross margin target even as electricity prices fluctuate. HaaS operating expenses (OPEX) include electricity, compression and purification, RO water consumables, and personnel costs estimated at $228,000 per year for a fully operational customer facility - these costs would be incurred by the consumer. Direct variable production costs are $3.96/kg of hydrogen, combining electricity, maintenance, and operational overheads.

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For the Electrolyzer Sales model, understanding per-unit CAPEX and OPEX clarifies how sales revenue translates into profit margins and reinvestment potential for future growth.
Each electrolyzer unit carries base equipment costs (COGS) of $15,000 for 25 kg/day, $150,000 for 250 kg/day, and $1.5 million for 2,500 kg/day systems. Installation, materials, personnel, and R&D costs are variable; they'll scale with production or customization level rather than fixed overhead. The model targets a 30% gross margin, reflecting the difference between these COGS and the selling prices listed in the revenue table ($19,500, $1.95M, and $2.535M, respectively). This cost structure enables Tobe to adjust margins per system size while maintaining predictable profitability as volumes increase.

HaaS Expenses		
H2 Cost	2.70	per kg
25kg Installation cost	1,500.00	per device
250kg Installation cost	15,000.00	per device
2500kg Installation cost	150,000.00	per device
Commision	0.18	per kg
Service cost	0.02	per device
Device Sales Expenses		
25kg Installation cost	1,500.00	per device
250kg Installation cost	15,000.00	per device
2500kg Installation cost	150,000.00	per device
25kg Commission cost	750.00	per device
250kg Commission cost	7,500.00	per device
2500kg Commission cost	75,000.00	per device
25kg Total cost	17,250.00	per device
250kg Total cost	240,000.00	per device
2500kg Total cost	1,725,000.00	per device

Table 3

Across both business lines, fixed costs consist of long-term CAPEX (plant construction, installed equipment, and depreciation), while variable costs include electricity, consumables, labor, and materials. So, in initial expansion stages, there are high fixed costs, but unit costs decrease as production scales. Under the HaaS model, OPEX is primarily variable, supporting scalability without major new investment per site. Under the Electrolyzer Sales model, costs remain flexible through contractor-based installation and third-party EPC channels. This cost structure makes Tobe's rollout scalable and capital-efficient, enabling expansion without proportionate increases in overhead.

5-Year Financial Projections

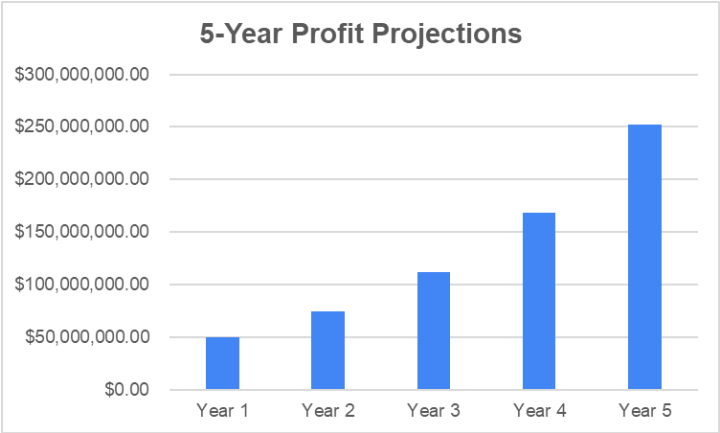

Figure 7

The financial projections for Tobe are built on a five-year horizon, beginning directly after the implementation of our proposed rollout plan. These projections model both revenue streams - Hydrogen-as-a-Service (HaaS) and Electrolyzer Sales - to reflect the company's hybrid market entry strategy. Each model is derived from benchmarked hydrogen pricing and equipment costs. For HaaS, revenues are tied to kgs of hydrogen produced and sold per day and for Electrolyzer Sales, revenues are based on unit sales of 25 kg/day, 250 kg/day, and 2,500 kg/day systems. Revenue models for both streams have an imposed profit margin of 30% because typically, the majority of electrolyzer manufacturers set their profit margin between 25% and 35% on system sales before scaling up production.

Growth rates assume a 10% annual success rate for pitched sales (given that this is just below typical for B2B sales), as well as a 10% market growth rate which reflects the compounded annual growth rate of the green hydrogen market.[45][46] The model emphasizes that Tobe's profitability can be achieved by scaling both service and equipment volumes, balancing long-term recurring income (HaaS) with short-cycle cash inflows (Electrolyzer-Sales).

Across both business lines, several common operating assumptions were applied to ensure consistency and comparability. Production sites under the HaaS model are assumed to operate 330 days per year to allow for scheduled maintenance and downtime (Clean Hydrogen provides a KPI assumption of 8000 working hours of a system which is about 333 days). Cost inputs -

[45] McGee, C., Average sales call conversion rate by industry: 2025 report, 2025
[46] Interview, Steve Syzmanski, Appendix E

including labor, materials, and O&M - are escalated at a 2.5% annual inflation rate, consistent with FED inflation targets.[47]

Threats & Opportunities Analysis

Government incentives and rising industrial demand are accelerating the shift toward green hydrogen, while regulatory uncertainty and infrastructure gaps require Tobe to scale carefully to capture opportunities.

Government policy creates several major opportunities for clean hydrogen, especially for smaller companies targeting rural or off-pipeline markets. Federal incentives, such as the Inflation Reduction Act's clean hydrogen production tax credits, significantly reduce the cost gap between green and grey hydrogen, making decentralized production more economically attractive. The Department of Energy and state-level programs also offer grants, rebates, and support for clean energy projects, while the $7 billion investment into Regional Hydrogen Hubs helps drive demand for new hydrogen technologies. These policies encourage industries like ammonia production, chemicals, metallurgy, and power generation to shift toward low-carbon fuels, creating a growing customer base. Rural energy and resiliency initiatives further support local, on-site hydrogen production, exactly the type of market segment that benefits from modular electrolyzer systems.

Similarly, federal agencies are building the domestic hydrogen supply chain. In March 2024, DOE awarded $750 million to expand electrolyzer R&D, manufacturing, and recycling across 52 projects. The DOE Loan Programs Office also issued a $1.66 billion loan guarantee to Plug Power for multiple green hydrogen plants, signaling that the federal government is ready to support first-of-a-kind production at scale. These moves are expected to reduce equipment costs, strengthen supply-chains, and expand the U.S. manufacturing base needed for widespread deployment.

At the same time, government-related threats create real challenges. Policy uncertainty around tax credit qualifications, renewable energy matching rules, and carbon intensity measurements could make compliance expensive for smaller producers. Water regulations also pose a risk, since electrolyzers require very clean water and many states face restrictions due to drought. Large companies such as Siemens, Plug, and Bloom have the scale and political influence to navigate shifting regulations more easily, creating competitive pressure. Other threats include slow development of hydrogen infrastructure, especially pipeline limitations due to hydrogen embrittlement, and the high cost of materials like platinum, which current electrolyzers rely on. Together, these factors make the market promising but also dependent on careful navigation of regulatory and technological risks.

[47] Inflation Targeting Explained: Central Bank Strategy for Price Stability

Reaching cost parity with gray hydrogen would be the green hydrogen industry's defining milestone. Today, gray hydrogen costs $1-2/kg, while green hydrogen costs between $4-8/kg depending on electricity prices, among other factors. Analysts forecast the price of green hydrogen to decrease by 40 to 80% between now and 2030[48], contingent on renewable power costs, electrolyzer scale-up, and continued policy support. The combination of 45V credits and cheap renewable power could bring effective green hydrogen costs down to $2–3/kg by the end of this decade. Tobe should time its large-scale HaaS rollouts to coincide with these falling cost curves by targeting anchor customers (refineries, fleet depots) by 2027–2030, when green hydrogen is projected to achieve economic viability without heavy reliance on subsidization.

Competitive Overview

By combining dual revenue models, low-cost green hydrogen production, and the ability to serve markets unreachable by legacy players or emerging alternatives, Tobe is positioned as the most flexible, affordable, and scalable solution for accelerating distributed hydrogen adoption.

Competitor & Criteria	Affordability	Clean Energy	Small-Scale Flexibility	Efficiency
Tobe Energy	✓	✓	✓	~ 94%
HyVera	✓	✓	✓	~ 90%
H_2	✓	✗	✗	~ 75%
Bloomenergy nel plug	✗	✓	✓	~ 60%
SIEMENS energy Rockwell Automation	✗	✗	✗	~ 81%

Figure 8

[48] Taibi, E., et al., *Green hydrogen cost reduction: Scaling up electrolysers to meet the 1.50C climate goal*, International Renewable Energy Agency, 2020

Tobe's dual-market strategy, direct electrolyzer sales and Hydrogen-as-a-Service, ensures revenue diversification while penetrating both capital-heavy and contract-based markets.

Tobe uniquely serves industries distant from hydrogen pipelines, including remote manufacturing, fleet transport, and data centers. Through its focus on small-to-mid scale adoption, data-backed operations, and zero-emission production, Tobe Energy stands out as the most affordable, flexible, and sustainable hydrogen solution currently available.

Tobe stands out from legacy players like Siemens, Rockwell, Bloom, Plug, and Nel because they are not tied to decades-old infrastructure, slow upgrade cycles, or the high-cost structures that come with traditional industrial companies and gray hydrogen producers. Instead, they are building a modern and modular electrolyzer designed from day one for lower cost per kg, higher efficiency, and real scalability. Siemens and Rockwell focus on broad industrial automation rather than hydrogen, while Bloom, Plug, and Nel often face high capital costs, complex maintenance requirements, and inconsistent uptime across different deployments. Tobe offers a simpler and more predictable system that replaces bulky and maintenance-heavy stacks with technology that is easier to deploy and service. Their approach is also designed to integrate seamlessly with partners such as Zeeco and Loves, allowing for immediate use in both grid and off-grid environments. As many competitors continue to struggle with the economics of green hydrogen and regularly miss their target production costs, Tobe is intentionally engineered to meet competitive cost goals without sacrificing reliability, giving them a strong advantage in new hydrogen markets that prioritize speed, affordability, and clean performance.

A competitor to Tobe that stands out is HyVera[49]. HyVera, a Canadian company that recently expanded to the US has two offices located in California and Indiana. HyVera, like Tobe, does not require expensive transportation which is one of the obstacles keeping hydrogen prices high. HyVera's technology produces fuel pellets that, when combined with water, results in a reaction that produces hydrogen. The leftover material can be reused a second time to produce more hydrogen[50].

[49] HyVera, "About HyVera Distributed Energy"
[50] Munz, "HyVera Announces US Launch."

Next Steps

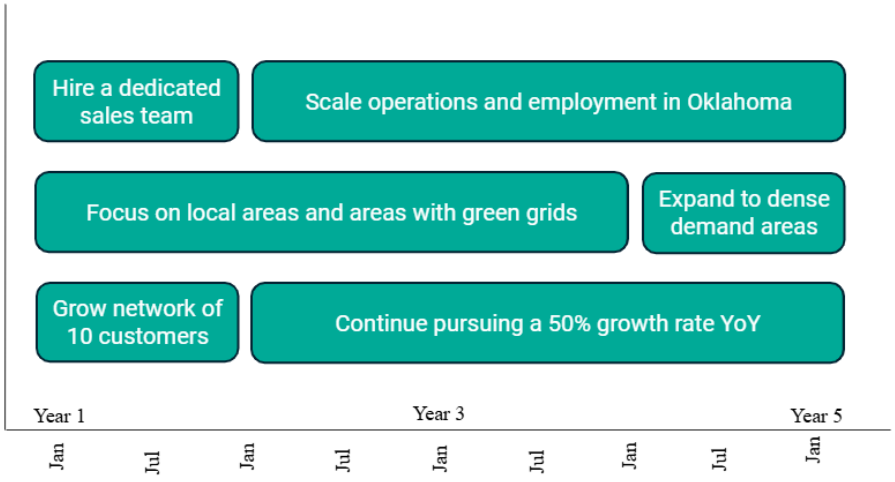

Figure 9

Network Development

- Build a network of 10 hydrogen users who are >150 miles outside of a hydrogen facility. Start with customers who are in or near Oklahoma to utilize existing connections.
    - Start with Oklahoma and surrounding states (OK/TX/KS/AR) to leverage existing professional connections and local hydrogen ecosystem partners.
    - Build a list of 100+ potential users using industry databases (D&B Hoovers, LinkedIn Sales Navigator).
    - Prioritize companies who meat ideal customer criteria.
    - Reach out with a standardized discovery script and schedule site visits.
    - Tap strategic partners for warm introductions, including:
        - Love's, Clean Energy Fuels, or regional fueling companies
        - Local chambers of commerce
        - Industrial parks & regional economic development authorities
    - Create a recurring virtual roundtable to engage prospects and establish credibility.
- Continue to grow Tobe's network by 50% every year.
    - Attend targeted industry events focused on Tobe's best-fit segments:
        - ACT Expo (transportation & depots)
        - Hydrogen Americas Summit
        - GlassBuild Expo (glass processing)

        - International Poultry & Processing Expo (food hydrogenation)
        - WEFTEC (wastewater treatment: biogas + hydrogen opportunities)
    - Form relationships with distribution channels that already reach industrial customers:
        - Industrial gas distributors (Airgas, Matheson)
        - EPC firms (engineering, procurement, construction)
        - Equipment suppliers serving mid-sized industrial facilities
- Hire a sales team with industrial equipment or energy sales experience to pursue leads. Each sales member should take on 10 clients a year, with a 10% success rate.
    - Equip salespeople with standardized tools:
        - Calculator for hydrogen delivered cost vs. onsite generation
        - Total Cost of Ownership comparison templates
    - Track KPIs:
        - of discovery calls
        - of feasibility studies
        - Close rate (target: 10%)
- Explore potential complementary partnerships to fulfill customer needs associated with hydrogen (ex: storage with Hexagon Purus, compression with Atlas Copco, renewable power with OG&E).

Operational

- Scale manufacturing operations to meet demand. Potential choke points to meet demand:
    - Skilled labor shortages – develop a training pipeline and partner with technical schools in Oklahoma City or Tulsa.
    - Long-lead-time components – pre-order long-lead items ahead of customer demand, lock in supply agreements, dual-source critical components where possible
- Hire an accountant to oversee all bookkeeping, financial statements, and regulatory filings

Legal and Security

- Continue developing security technology to protect Tobe intellectual property.
- Hire a law firm to manage Tobe customer contracts and liability with both sales models.

Long-term Opportunities:

After establishing capital and routine success, explore other business opportunities such as:

- Manufacturing facilities in other states where HaaS demand is highest

- International markets such as Europe and Southeast Asia
- Tobe owned and operated fueling stations

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Appendix

Appendix A

Email answers from Amish Patel, Business Solutions, Enterprise Products

What are the operations behind Hydrogen production?

> The primary source of Industrial Hydrogen is from Natural Gas (CH4). Hydrogen is a product of the SynGas (Synthesis Gas) process, which is decomposing hydrocarbons into hydrogen and carbon monoxide gases. The most energy efficient way to create pure hydrogen is "reforming" natural gas to H2 and CO.
>
> Another industrial way of making hydrogen is via electrolysis, which is running electricity through water (adding electrolytes improves the reaction) with a cathode (where the H2 forms) and anode (where the O2 forms). It requires a lot of power and is very corrosive on the electrodes. However, if you have a green energy source (wind, solar, etc..), it is the only "green" way to produce hydrogen, so the renewed interest recently.

How do Hydrogen contracts work?

> I have not worked with the contracts too much, but as much of the oil and gas/chemicals industry requires, a long-term deal is needed to support the large capital investment required to build a hydrogen plant and associated pipeline. The key players in the hydrogen business are the Industrial Gas companies (Air Liquide, Air Products, and Linde are the major ones in the US). So, the customer(s) will typically pay a fee to consume or will sell hydrogen to these Industrial Gas companies.

How is Hydrogen moved?

> The bulk of hydrogen is moved by pipeline. Additionally, hydrogen can be moved on multi-cylinder trailers or liquified bullet trailers by semi-trucks. Cylinders are also delivered as you might find in the labs at OU.

What does Hydrogen end use look like?

> Bulk usage is for chemical processes as a feedstock. In the refining industry, it is used to remove sulfur from fuels. Sometimes it is used as fuel in facilities since it requires further processing to purify. Very little is used for powering vehicles.

How is green hydrogen changing the industry?

> Predominantly hydrogen is used to hydrotreat fuels in refineries, reducing drastically the amount of sulfuric compounds in fuels, which when burned can produce SO2 and SO3 that will turn in the atmosphere by mixing with water to making acid rain (H2SO4 – sulfuric acid). Hydrogen has tried to be commercialized as a fuel source for vehicles, but the market has not supported it.

Appendix B

Interview with Amish Patel, Business Solutions, Enterprise Products

What are factors in contracts

Industrial customers primarily use hydrogen for chemical processes. These users typically rely on long-term contracts due to the high capital expenses associated with production and infrastructure. Natural gas feedstock costs can reach approximately $300 million, and small-diameter hydrogen pipelines cost $3–4 million per mile.

Fee-based contracts are common; fee levels correlate directly with hydrogen production costs, averaging roughly $0.70 per kg. Companies target a 12–15% hurdle rate above cost to achieve acceptable profitability. Anchor customers are a critical requirement for project viability.

The hydrogen market is dominated by major players including Air Liquide, Linde, Air Products, Nippon Steel, and Messer (with Messer absorbed by Air Liquide in the U.S.). Hydrogen transportation by truck is limited to roughly 1,800 lbs. in gaseous form. Only two plants currently support liquid hydrogen transport (requires verification). Smaller customers, such as school laboratories, purchase hydrogen in very small quantities.

Where are H pipelines/refineries?

On-site hydrogen production provides substantial operational value. Hydrogen generation produces multiple by-products and requires significant heat inputs; these outputs must be effectively managed. Integrating production within a refinery enables the steam by-product to be absorbed and reduces environmental imbalance.

Large-scale production benefits from economies of scale to achieve acceptable energy efficiency. Existing hydrogen pipeline infrastructure in the U.S. is concentrated along the coast, particularly between Corpus Christi and New Orleans. There are three underground storage wells along these pipelines, with a possible fourth in development. Another potential storage project exists in Utah, tied to an electrolysis and long-duration storage concept.

Pipeline routing and system information can be found on operator websites and through mapping resources maintained by the Texas Railroad Commission. Globally, the largest hydrogen pipeline systems are expected to be in the United States, Europe, Saudi Arabia, and China. Tax incentives encourage placing hydrogen plants within refineries and moving hydrogen by pipeline. Upfront costs include plant construction followed by customer-specific pipeline investment.

What kind of niche or smaller industries is hydrogen moving into?

Hydrogen currently supports various chemical reaction processes, although this sector is not experiencing significant growth. Enterprise-level users rely on hydrogen for these applications. The strongest growth area remains hydrogen-fueled vehicles, particularly in California. Japan is utilizing hydrogen for ammonia production. Hydrogenation markets are not expected to expand, especially with political factors such as RFK's health initiatives.

The electronics and semiconductor sector represents a growing but small-scale use case; as chip manufacturing expands, hydrogen demand increases accordingly. Certain automotive manufacturers, such as Honda, demonstrate potential synergies between hydrogen and electric vehicles. Hydrogen may also see niche residential adoption for hydrogen-powered cars. Opportunities exist in energy generation and in server applications (e.g., Bloom).

What is the difference between small and large electrolyzers?

Electrolyzers require substantial electricity. Small electrolyzers can operate effectively because their power demands remain manageable. Large-scale electrolyzers, however, require extremely high electricity input, raising concerns about sourcing adequate and reliable power. In Texas, the ERCOT grid provides a mixed energy supply but experiences measurable power loss. Both grey and green hydrogen production at scale involve trade-offs related to electricity requirements and efficiency.

Are there any markets you see transportation costs being a limiter?

Transportation costs can limit hydrogen market expansion. Historically, hydrogen growth has been tied to cleaner fuel applications. Fuel cells present a possible pathway for residential power. However, regions highly sensitive to CO_2 emissions are generally unsuitable for hydrogen production due to process-related impacts.

What would you say is the most important for H use: reliability, price, sustainability?

The hydrogen market presents a unique dynamic. Although hydrogen is marketed as a green energy solution due to its lack of downstream pollutants, industrial buyers prioritize cost above sustainability or reliability, provided production does not release harmful toxins. The environmental benefit of hydrogen depends significantly on the energy source used during production.

Hydrogen remains a low-margin industry, where profitability depends heavily on volume. A key recommendation is to review the historical growth trajectory of hydrogen to understand long-term market behavior.

Amish's advice for us was to look at how hydrogen has grown historically and look for the indicators for where it will grow next.

Appendix C

Interview with Ashish Bhakta, Alternative Energy R&D, Love's Travel Stops

Key factors of a contract to buy Hydrogen Fuel from Producer:

- Term
- Volume commitment
- Price- fixed or variable, variable preferred
    - Index around price of producing the commodity and overhead
- Middlemen involved (supplier, distributor, end customer)
- How a public client (like buses) does this
    1. Request for Proposal- asking for bids is required when spending public dollars
    2. Contractors (like Love's) bid on RFP w/ designs and costs
    3. Maintenance and other on-going deals are ironed out once the initial implementation is done
    4. Contractors buy the equipment from suppliers (like NEL) and Love's sells hydrogen to their customers for projects
- How to for private company
    1. Starts w/ customers desire (i.e., building an H2 station at an existing Love's)
    2. Work out cost/benefit on if/where it makes sense to invest in building
    3. Capital costs as well as cost of procuring fuel and maintaining facilities

Love's Hydrogen Facilities are mostly for fueling buses currently

Few suppliers mean little price competition – **Suppliers have pricing power, not buyer –**

Region, production method, and H2 phase (gas vs liquid) all have large impacts on price

Love's has ~7-8 distribution centers currently

Half receive and distribute Liquid H2, other half receive and distribute gaseous

Liquidizing requires large scale (>30,000kg)

Can Tobe's devices produce enough H2 to condense into liquid? Might limit users

One tank of liquid H2 is ~10x denser for transport

At certain scales, buying liquid H2 makes more sense than running a plant to manufacturer it

Being able to make your own fuel can be empowering and impactful for the first time in history

Current administration has significantly slowed down interest across the board

"...in the past 4 years, everything was peaking, idk how to even describe it. But since the end of 2024, it's slowed down, except for the West Coast… California is still trying to be aggressive."

Mandates in some states require 0 emission buses by certain date- **ICT rule**

Potential Competitors:

Clean Energy- main competitor, CNG, LNG, RNG, H2 fueling

Hydrogen IDCs like Linde, a legacy H2 producing company that wants to provide directly

Time to build facility depends on their needs- all of these are long term builds

- Can take years to even confirm whether a build will happen
- Public projects are more predictable but take ~1 year for project to even start project
- Private projects have more open-ended timetables, but also more cases of urgency
- Strictly facility side can be built in a number of months
    - Get request w/ volume commitment
    - Find supplier in the area that fits
    - Being buying equipment

Typical H2 fuel contract is 3-5 years currently Not very long compared to traditional fuel and costs of these projects.

- Current customers do not want to lock themselves, potentially due to:
    - Political changes
    - Riskier new commodity

Middlemen, like Love's, want to be involved in all sides of energy and fuel supply industries

For bigger or more specialized projects, Love's or developer can buy from more specialized suppliers **(like Tobe?)**

Love's constantly juggling new suppliers

Love's constantly looking for new alternatives for equipment they sell to their customers

Bounce between lots of suppliers, like Bloom, Nel, etc.

Love's typical production on site is 1MWatt electrolysis which makes ~430 kg/day

One bus use ~20-25 kg per day, with each customer having 100+busses

Liquid is better for scaling and fueling, but more expensive to produce w/o large scale

Producing their own fuel requires tons of infrastructure–

Ideologically motivated customer

Love's customer facility near University of Illinois- Customer didn't feel comfortable w/ supplier market, they wanted their H2 to be fully renewable (not common, especially with new administration), and geography can impact decision. Now run their own 1MW electrolyzer + 2MW Solar array to power facility making fully renewable H2 for their bus fleet. Got great PR but was very expensive project- chose to do this for the mission over the money.

Appendix D

Email response Ashish Bhakta, Alternative Energy R&D, Love's Travel Stops

1. What price changes could a growth in efficiency cause?

I don't think I can answer this question as it's contingent upon too many variables. Are you talking about the entire value chain or something more specific like, electrolyzer efficiency?

2. How many electrolyzers are Love's buying from suppliers on average (per project)?

Each project we have or will have 1 electrolyzer.

3. Are Love's competitors bigger, smaller, or the same? (this is to develop a normalized anecdotal #)

Love's is a privately held US based company, but I think you can look up financials on Forbes. I would say the Industrial Gas Companies (IGCs) are global companies and are likely much larger.

4. Could you describe the primary supply-chain and operational vulnerabilities for hydrogen projects? (e.g., transport, storage, skilled labor)

Due to the nascency of the market, it's all of the above. Production or limited number of sites with production, transport or timely delivery, specialized labor to build and maintain the sites, etc.

5. Where do you think hydrogen demand will grow fastest in the next few years?

For transportation, it will be based on where the market is subsidized the heaviest as that's the only way it can compete with alternatives.

Or markets like California, where there are certain mandates that push end users to transition.

You also mentioned constantly searching for new suppliers based on efficiency. During my research, I came across a relatively new electrolysis company that reports operating efficiency around 94.7%. Here's their website in case it's of interest: www.tobe.energy [tobe.energy].

Thanks, I'll take a peek! As you can imagine, 2020-2024 was a booming time for new hydrogen technology companies.

If you know anyone else I could speak to that would be able to round out my assessment of the green hydrogen market, please let me know!

I do have a contact from an IGC partner, though they won't have much to share specifically on green hydrogen. Please let me know if you would like me to connect you.

Appendix E

Interview with Steve Szymanski, Vice President of Sales, and Marketing, Nel

How much does an electrolyzer cost?

The cost of an electrolyzer itself is relatively small compared to the total cost of producing hydrogen. Approximately 70% of total production cost comes from electricity, underscoring the importance of efficiency. Water and capital expenses contribute minimally to overall cost. As a result, cheap power directly translates into cheap hydrogen, and access to green energy is essential; without renewable inputs, the environmental value proposition of hydrogen is diminished.

A major industry benchmark is the goal of building large-scale electrolyzer plants for $1,000 per kW, a price point regarded as highly attractive. Utilities are treated strictly as operating costs, as electrolyzer manufacturers do not produce hydrogen, they only sell equipment. In all cases, the plant owner, not the equipment company, is responsible for utility expenditures.

"If you can build an electrolyzer plant (large scale) for $1000/kW a lot of people would be really excited about that."

When you approach customers with assumptions, how do you sell? How do you find them?

When approaching customers, assumptions about costs and operating conditions influence how the value proposition is communicated. In an unsubsidized market, grey hydrogen remains cheaper than green hydrogen. Bridging this cost gap requires demand-side incentives, such as pay-for-difference programs or carbon taxes. While many companies pursue ESG goals, the cost differential between green and conventional hydrogen remains challenging to justify commercially.

Can you describe some of the markets you have seen reach out to Nel?

Companies reaching out to electrolyzer suppliers tend to be conventional hydrogen users, particularly those who rely on hydrogen as an input chemical. Typical sectors include semiconductor manufacturing, steel treatment, and glass processing. Demand for green hydrogen within these traditional industries is growing, and these markets remain core opportunities for equipment providers.

Customers frequently evaluate whether it makes more sense to receive delivered hydrogen or produce hydrogen on-site. Many prefer self-reliance and are hesitant to enter long-term supply contracts. While industrial decarbonization attracts more public excitement, traditional markets still represent steady, reliable demand.

Have you noticed that customers are far away? What's the main motivation?

Transportation logistics strongly influence customer decisions. Compressed hydrogen has a limited transportation radius of roughly 150 miles, making it impractical for long-distance delivery. Liquid hydrogen travels further but is more expensive and susceptible to boil-off losses. In some regions, hydrogen is delivered from Canada, especially along the East Coast.

The guiding principle is that "the best place to put an electrolyzer is at the point of use." This reduces transportation costs and dependency on delivered fuel.

Central vs decentralized plant?

Certain customers prefer fully owning and operating their electrolyzer systems. For example, a steel manufacturer in West Virginia purchased equipment to remain self-sufficient.

A notable example of centralized production is a planned 60-ton/day electrolyzer and liquefaction plant in Ardmore, developed by Woodside Energy. The plant would sell hydrogen to companies fueling trucks, but the project is currently delayed due to failures among hydrogen truck startups. Many customers in similar projects are large oil and gas companies who view hydrogen as an additional project type and a pathway into decarbonization markets.

Would HaaS be a feasible model?

Hydrogen-as-a-Service (HaaS) models carry significant risk. Cost exposure is a key concern: if hydrogen is sold for $5/kg but actual production costs land at $6/kg, the provider incurs direct losses. Manufacturers often consider it "cleaner" to simply sell electrolyzer equipment rather than sell hydrogen itself.

Leasing models have been explored since leasing is categorized as an operating expense for customers. However, categorizing the sale of hydrogen molecules within such a model introduces complications.

Industrial gas suppliers often impose penalties and force majeure terms that place heavy burdens on customers, reinforcing the appeal of customer-owned systems. Even with on-premise hydrogen generation, selling hydrogen to the customer still creates exposure similar to traditional gas supply models.

How do contracts work?

Electrolyzer sales typically follow a structured payment schedule. Once equipment is installed, manufacturers require full payment. To maintain long-term engagement and recurring revenue, suppliers also provide long-term service agreements, covering maintenance and support for the installed equipment.

Appendix F

Email correspondence with Steve Szymanski, VP of Sales, and Marketing, Nel

The length of the sales cycle and the "win rate" varies widely depending on the size of the electrolyzer system. For example, we have some smaller machines (just a few kW of input power) that we can sometimes sell in just a few months from our initial engagement with the customer. We also don't have a lot of competition in that size range, which helps. As I was telling you in our call, the real competition is delivered hydrogen. So, if a customer has already made up their mind that they want to produce their own hydrogen, our win rate is quite good. Even so, these are still capital purchases, as even the smaller systems cost $100K. Customers typically do capital budgeting on an annual basis. So, if you start talking to them in 2025, and they don't have money approved in their 2025 capital budget, they will end up deferring the purchase to 2026. Some customers prefer to pay for hydrogen as an operating expense, and that is where delivered hydrogen is their preferred solution.

For larger electrolyzer equipment (MWs of power), the sales cycle can be very long (2+ years in some cases), and many projects don't end up happening because the client can't get financing. In these types of projects, the success rate can be quite low, even though our equipment is preferred by the customer. I can't put a specific number on it, but anecdotally I would guess that less than 10% of larger projects we quote end up turning into a sale of some kind. Again, these are quotes

for millions of dollars' worth of electrolyzer equipment, and they will often require significant third-party financing. The low success rate is more a function of the project not being "bankable" than the client not liking our technical solution. Hopefully, that makes sense.

Appendix G

Interview with Dipankar Ghosh, ConocoPhillips Chair in Energy Accounting, University of Oklahoma

According to Dipankar Ghosh, the cost of green hydrogen is currently not viable. The central issue is that electrolyzers remain too expensive, though companies like Tobe aim to reduce this barrier. Green hydrogen's competitiveness relies heavily on subsidies, which play a major role across the hydrogen industry.

A recurring theme is the importance of lowering electricity costs. Cheap electricity directly reduces hydrogen prices, making overall production more feasible. Companies often use excess electricity—particularly medium-sized power plants—to generate hydrogen. Storage represents another significant cost, and reducing the cost of green electricity is viewed as essential to improving market viability.

The primary question companies ask is: "How much carbon are you releasing?" Environmental impact is central to green hydrogen adoption. Many companies are adopting CO_2 reduction goals, and consumer awareness is driving interest in greener alternatives.

Green hydrogen's output is oriented toward green or low-carbon applications only, limiting its market reach compared to grey hydrogen. India is currently the largest investor in green hydrogen, and global adoption is growing more rapidly outside the United States.

Transporting hydrogen is both cost-intensive and logistically complex. This serves as one of the hidden and often underestimated costs of hydrogen supply. Storage adds further expense, reinforcing the need to locate hydrogen production near the point of use.

Dipankar noted meaningful potential for on-site electrolyzers, particularly for smaller businesses. On-site generation helps avoid transportation challenges and reduces exposure to hidden costs associated with delivery and storage. Certain sectors stand to benefit from on-site production, especially those prioritizing self-reliance and sustainability.

There is more global growth in green hydrogen than in the U.S., driven by national sustainability targets and consumer expectations around environmental impact. The industry's long-term trajectory will depend heavily on electricity affordability, continued subsidies, and companies' ability to meet their green and CO_2-based performance goals.

Appendix H

Interview Notes with Dr. Runar Nygaard, University of Oklahoma

Hydrogen

It's important to know that hydrogen is not an energy source, it's an energy carrier. It can also be used for storing energy. On-site production could be better if we know how to produce on-site, but current production is large-scale and industrial complex.

Hydrogen Pipelines

There are researchers at OU that are looking at how much hydrogen can be added into ordinary pipelines without having to build a new pipeline system. There are companies right now that are adding about 5% of hydrogen into their natural gas stream. They think that up to 20% of hydrogen could be added into natural gas pipelines but they don't know and there are no standards yet. He said that the question is whether it might be better to take hydrogen and add carbon dioxide to it to create a synthetic fuel. He said it's much more like natural gas or gasoline; it looks like a hydrocarbon which makes it easier to transport.

The current hydrogen pipelines go between manufacturers to other petrochemical plants that use hydrogen. He explained that if we all go in for hydrogen, the current pipelines would need to be replaced. However, current pipelines could be improved by putting liners in every pipeline. The reason hydrogen can't be transported in natural gas pipelines is because the stainless steel used for natural gas pipelines is not resistant enough for hydrogen. Hydrogen causes the stainless steel to become brittle and the gaskets that are made from rubber lose their elasticity.

Hydrogen and location

He said that hydrogen uses a lot of fresh water. A lot of places where hydrogen can be beneficial are in areas with excess electricity production, e.g. Oklahoma, Texas, and Western USA. Such areas have a lot of wind and solar power which already uses a lot of water. Thus, it will create a strain on the water supply. He explained there's a mismatch between where the cheap and excess electricity is and where it would be best to produce hydrogen. Producing hydrogen in the Great Lakes area would be good considering the amount of fresh water.

Appendix I

Interview with Lance Lobban, Professor of Chemical Engineering, University of Oklahoma

1. Cleanliness and Cost of Hydrogen Pathways

Professor Lobban emphasized that green hydrogen is only clean when powered by solar or wind electricity. Grey hydrogen remains significantly cheaper because it relies on natural gas (methane) as its feedstock. Transitioning to green hydrogen requires clean water, excess renewable electricity, and customers who prioritize low emissions.

Electrolysis remains a major inefficiency in the hydrogen supply chain. Electrolyzers also require very clean water, adding an additional cost burden. High-pressure electrolysis could help with storage needs but would make equipment more expensive.

2. Transportation and On-Site Production

Transporting hydrogen is expensive, making on-site production especially valuable—particularly in rural or off-pipeline regions, which align well with your target market. Hydrogen pipelines cannot simply be repurposed because the gas causes metal embrittlement, which compromises pipeline integrity.

Hydrogen can be stored chemically in liquids, but this approach requires large chemical plants and results in energy loss during each conversion step.

3. Storage Challenges

Storage remains one of the largest unknowns and most significant challenges for the hydrogen industry. Professor Lobban noted that batteries make more sense for small-scale storage, while hydrogen is more viable for mid- to large-scale energy storage applications.

4. Industrial Users and Applications

The largest current users of hydrogen include:

- Ammonia production
- Metallurgy
- Chemical manufacturing
- High-heat industrial processes
- Power generation via fuel cells

Hydrogen for vehicles is technically feasible, but costly, because fueling systems and gas tanks require redesign.

5. Market Fit and Customer Considerations

For smaller, rural customers with modest hydrogen demand and limited pipeline access, on-site electrolyzer solutions are practical and logical. Leasing models—like solar panel leasing—may accelerate adoption by lowering upfront costs and allowing technology upgrades over time. Selling machines outright risks locking customers into outdated technology.

Selling hydrogen directly is less attractive due to the high cost of transportation.

6. Technology, Materials, and Industry Structure

Electrolyzers often rely on platinum catalysts, increasing capital cost; research into cheaper alternatives is ongoing. High-pressure electrolyzers may aid storage but would bring higher upfront equipment costs.

Industry structure is likely to favor large companies such as Siemens, which have the scale to integrate production systems, fuel cells, and equipment. Smaller companies (e.g., Plug, Bloom) may remain niche players or become acquisition targets.

Appendix J

Interview with Louis Mousney, Co-founder of Tobe

1. Background and Expertise

Louis Mousney is based in Paris and currently works at Arkema, a major French chemical company. He leads the firm's green hydrogen strategy, covering both technical and commercial dimensions. He possesses extensive knowledge across the entire hydrogen value chain, including global markets, policies, electrolyzer manufacturers, production economics, transportation, storage, and industrial end-users.

He recently submitted his three-month notice to leave Arkema and will join Tobe Energy full-time as CFO and Co-Founder alongside Colby. Mounsey has been involved with Tobe since its early days due to his close relationship with Colby.

Tobe's near-term focus is the U.S. domestic hydrogen market, with no immediate plans for international expansion.

2. Overview of the U.S. Hydrogen Market

The U.S. hydrogen market is currently 95–99% gray hydrogen, produced via steam methane reforming (SMR). Green hydrogen adoption is limited almost entirely by the high cost of renewable electricity and the high cost of hydrogen transportation and storage.

Hydrogen is extremely expensive to move. For example:

- Production cost may be ~$3/kg
- Delivered cost in places like Tulsa or Seattle can reach $25/kg due to compression, trucking, and handling

This transportation premium is one of the biggest structural inefficiencies in the market.

3. Tobe Energy's Value Proposition

Tobe's model centers on placing electrolyzers directly at the customer's site, eliminating transportation costs. This allows Tobe to be cost-competitive even without subsidies.

Ideal initial markets include regions with:

- High industrial hydrogen usage
- No local hydrogen production
- High delivered hydrogen prices
  Examples include Tulsa and Seattle.

4. Partnership With Zeeco

Zeeco is a major global player in combustion engineering and testing, working with clients such as Chevron to develop hydrogen-capable burners. Their testing facilities handle:

- 100% hydrogen combustion
- 100% natural gas
- Mixed hydrogen/gas blends

Their customers seek flexible burners capable of running on multiple fuels.

Why Zeeco Needs Hydrogen

Zeeco already consumes hydrogen for burner R&D. Currently, they source hydrogen via tube trailers from Houston, which is expensive.

Tobe will replace this supply with on-site green hydrogen produced using their electrolyzer.

Simplified Supply Chain for the Partnership

1. Tobe installs electrolyzer at Zeeco's site.
2. Zeco provides renewable electricity to power it.
3. Electrolyzer splits water to produce green hydrogen.
4. Hydrogen is used directly on-site in combustion systems.

5. No downstream distribution is needed.

Financial Mechanics

- Zeeco currently buys delivered gray hydrogen at high cost.
- With Tobe, they stop buying delivered hydrogen.
- Instead, they pay Tobe for on-site green hydrogen or enter a structured partnership.
- Zeeco lowers operational expenses, while Tobe remains price competitive.

Why Zeeco Is Considered a “Partner”

Zeeco positions itself as a partner to its clients. For Tobe, this partnership provides:

- Industry visibility
- Credibility and validation
- A major testing environment
- Access to clients who need hydrogen test infrastructure

5. Business Models: Equipment Sales vs. HaaS

1. Equipment Sales (Electrolyzer Sales)

- Tobe sells the electrolyzer directly.
- Customer owns, operates, and maintains it.
- Lower risk for Tobe; simpler contracting.
- Revenue is traditional and upfront.

2. Hydrogen-as-a-Service (HaaS)

- Tobe owns and operates the electrolyzer.
- Tobe sells hydrogen generated on-site.
- Recurring, higher-upside revenue model.
- Significantly higher operational and financial risk.

Risks and Constraints of HaaS

- Liability for leaks, explosions, and safety incidents
- Compliance obligations: EPA, DOT, and permitting
- Operational staffing and monitoring needs
- Need for external financing (banks, investors)
- Revenue exposure to hydrogen price fluctuations
- Customer demand risk if usage drops

Because of these risks, HaaS requires careful contract design.

6. Contract Length Expectations

Industry-standard hydrogen contracts run 3–5 years, regardless of green or gray hydrogen. Customers hesitate to sign longer agreements due to emerging technology uncertainty.

Traditional SMR plants require long-term contracts because their financing depends on predictable offtake. Tobe's modular electrolyzers do not require long financing cycles, so contract lengths can remain flexible or shorter.

7. International vs. U.S. Market Differences

Factors That *Do* Translate Internationally

- Hydrogen cost drivers: electricity, electrolyzer capex, O&M
- The global need to reduce hydrogen cost
- Major market players (Air Liquide, Linde, Air Products) operate worldwide
- End-uses and technology trends are globally similar

Factors That *Do Not* Translate

- Policy stability
    - The U.S. has volatile hydrogen policy (administration changes)
    - The IRA sunsets in 2026
    - Europe offers long-term, stable support

- China and the Middle East follow predictable state-driven roadmaps

U.S. policy uncertainty is a major barrier to long-term project planning.

8. Cost Modeling Guidance

Louis recommends modeling hydrogen cost using the following drivers:

- Renewable electricity cost (largest factor)
- Electrolyzer capital cost
- Operations & maintenance
- Purification requirements
- Compression needs
- Storage needs

Your model should include:

- A with-subsidies case (IRA + state incentives)
- A no-subsidy case (post-IRA)
- Regional consumption adjustments by industry density
- Comparison of delivered gray hydrogen prices to demonstrate savings

9. Hydrogen Use in Data Centers

Hydrogen is emerging but not yet widespread in data centers. Current applications include:

- Backup power
- Peak shaving to avoid expensive daytime electricity

Adoption is slowed by storage cost, and commercial-scale implementation is not yet common.

Appendix K

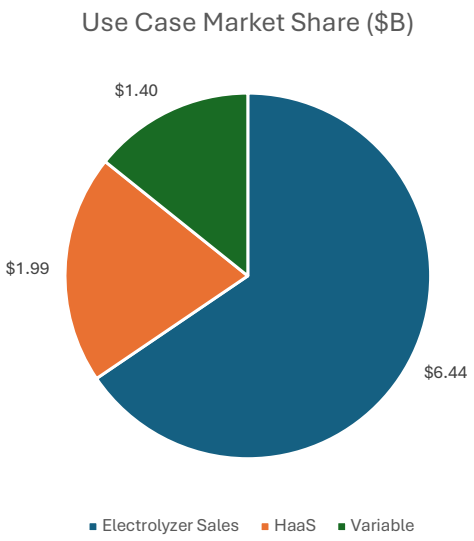
Ideal Customer	**Location**	**Segment**
RTC Washoe	Washoe County, NV	2. Urban Fuel Fleeting
Green Bay Metro Transit	Green Bay, WI	2. Urban Fuel Fleeting

CF Industries	Sergeant Bluff, IA Claremore, OK Yazoo City, MS	5. Chemical production & synthetic fuels
MBTA	Boston, MA	2. Urban Fuel Fleeting
Charlottesville Area Transit	Charlottesville, VA	2. Urban Fuel Fleeting
FreezPak	Suffolk, VA Miami & Jacksonville, FL	3. Light & medium-duty commercial fueling
BMW Manufacturing	Multiple locations, On map: Spartanburg, SC	3. Light & medium-duty commercial fueling
Comet	Columbia, SC	2. Urban Fuel Fleeting
United Natural Foods, Inc.	Multiple locations, On map: Sarasota, Florida	3. Light & medium-duty commercial fueling
Tyson Foods	Multiple locations, On map: Springdale, AR	3. Light & medium-duty commercial fueling 4. Industrial decarbonization
Walmart Distribution Centers	Multiple locations, On map: Arizona, Colorado, Nebraska, Wyoming	3. Light & medium-duty commercial fueling
FedEx Freight	Multiple locations, On map: Missoula, MT	3. Light & medium-duty commercial fueling
Uline	Multiple locations, On map: Atlanta, GA	3. Light & medium-duty commercial fueling
Landus & TalusAg	Boone, IA	5. Chemical production & synthetic fuels
Yara	Tampa, FL	5. Chemical production & synthetic fuels
Anheuser-Busch	Multiple locations, On map: St. Louis, MO	3. Light & medium-duty commercial fueling
Microsoft Data Centers	Multiple locations, On map: Arizona, North Carolina, Georgia	6. Backup power & microgrids
CT Transit	Hartford, New Haven, Stamford, CT	2. Urban Fuel Fleeting

Companies whose customers are ideal Tobe customers	**Location**	**Segment**
Ballard Power Systems	Bend, OR	6. Backup power & microgrids
Bobcat	West Fargo, ND	3. Light & medium-duty commercial fueling
Nikola	Pheonix, AZ	2. Urban Fuel Fleeting

ECL Data Centers	Mountain View, CA	6. Backup power & microgrids
New Way	Scranton, IA	3. Light & medium-duty commercial fueling
Generac Power Systems	Waukesha, WI	6. Backup power & microgrids
Nuvera	Billerica, MA	6. Backup power & microgrids
Hyundai	Bryan County, GA	2. Urban Fuel Fleeting

Figure 1



Segments	Market Share	H2 Need	Established Infrastructure	Sales Model
• Heavy-duty fueling • Industrial decarbonization • Chemical production & synthetic fuel	6.44 B	High	High	Electrolyzer Sales
• Light & medium-duty commercial fueling • Backup power & microgrids • Energy storage	1.99 B	Medium	Low	HaaS
• Urban fleet fueling • Pre-existing green companies	1.4 B	Variable	Variable	Variable

Figure 2

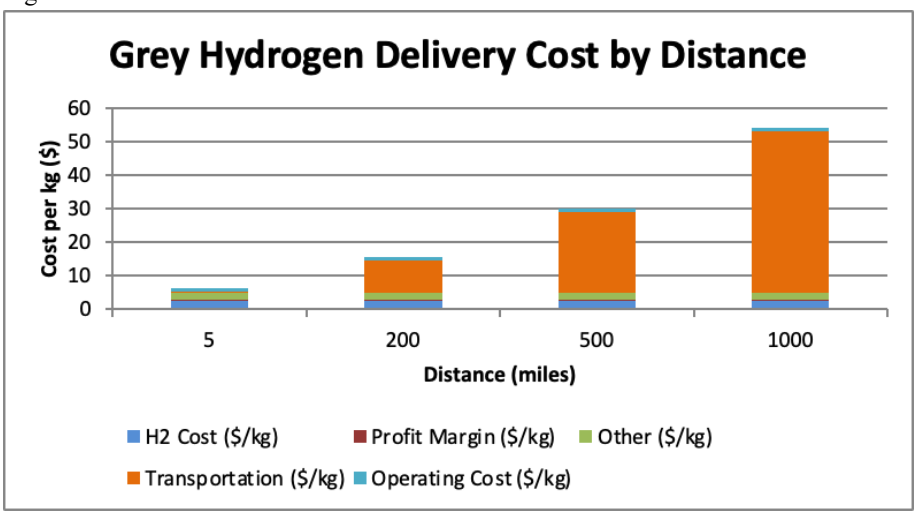
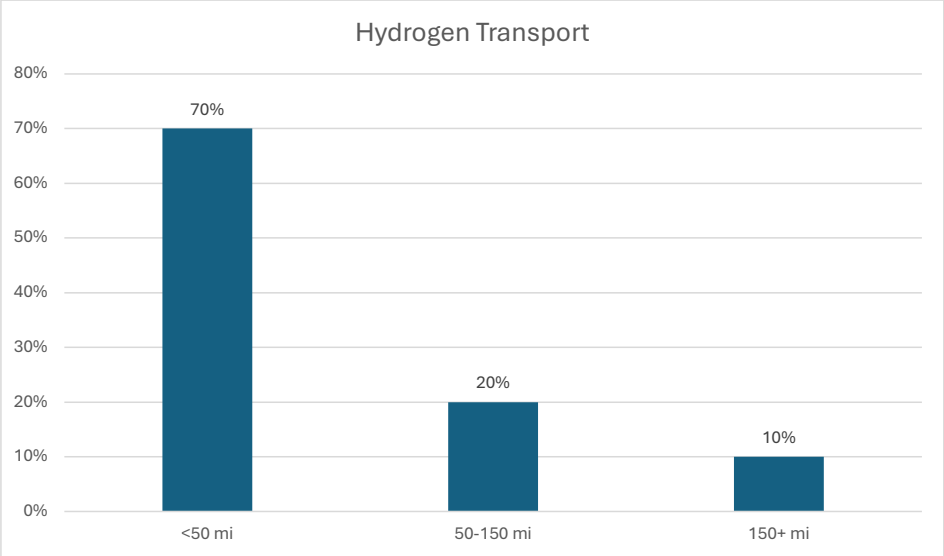


Figure 3



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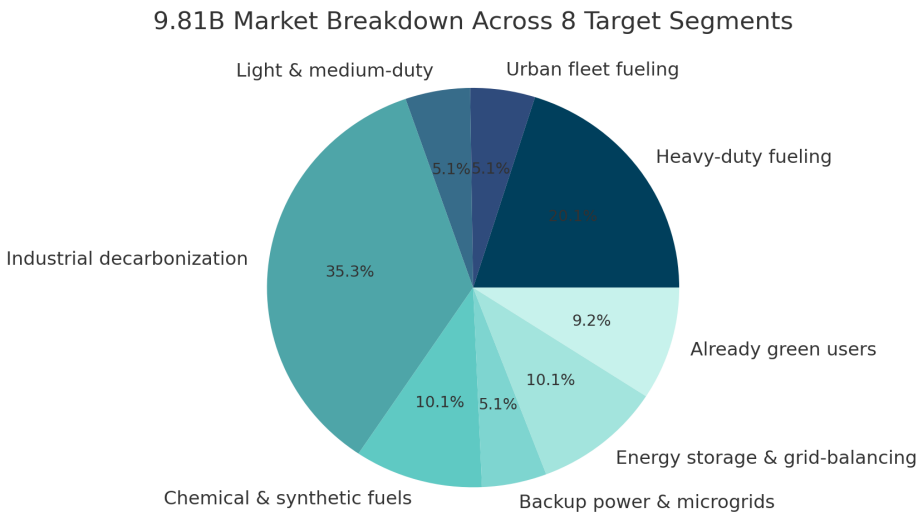

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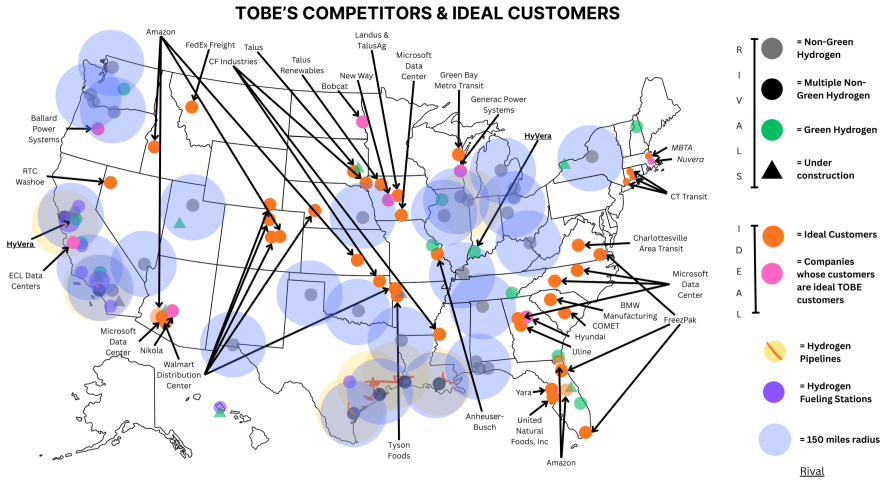

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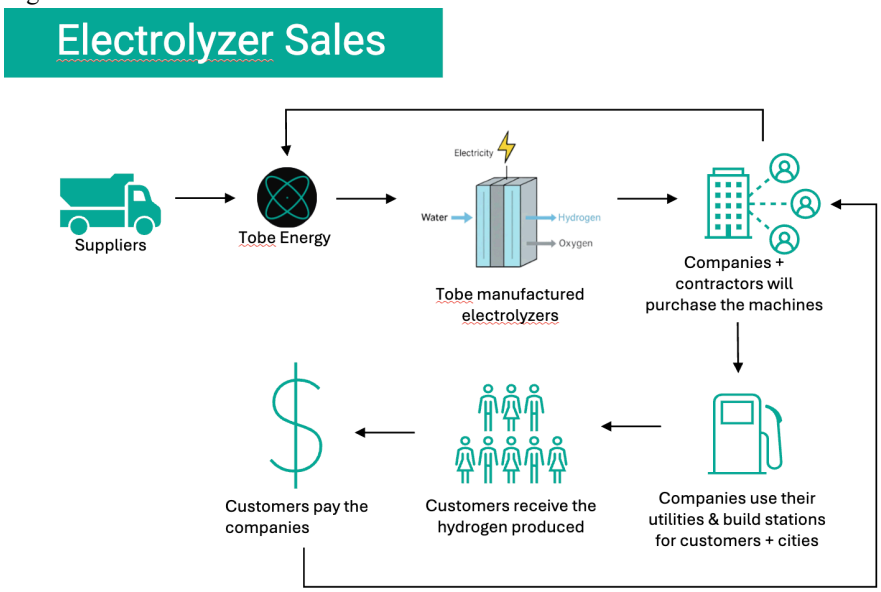

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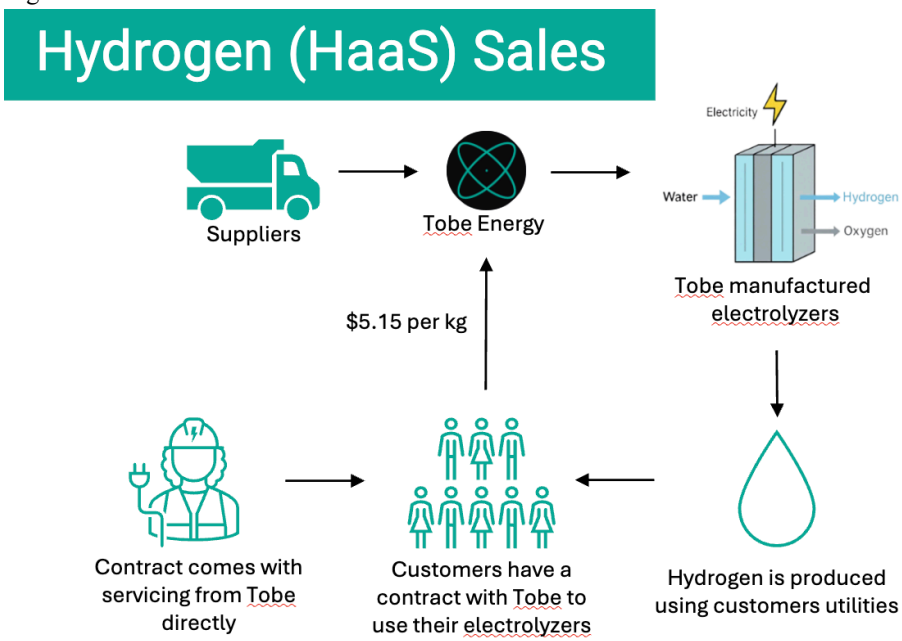
Hydrogen (HaaS) Sales
Suppliers
Tobe Energy
Electricity
Water
Hydrogen
Oxygen
Tobe manufactured electrolyzers
$5.15 per kg
Contract comes with servicing from Tobe directly
Customers have a contract with Tobe to use their electrolyzers
Hydrogen is produced using customers utilities

Figure 8

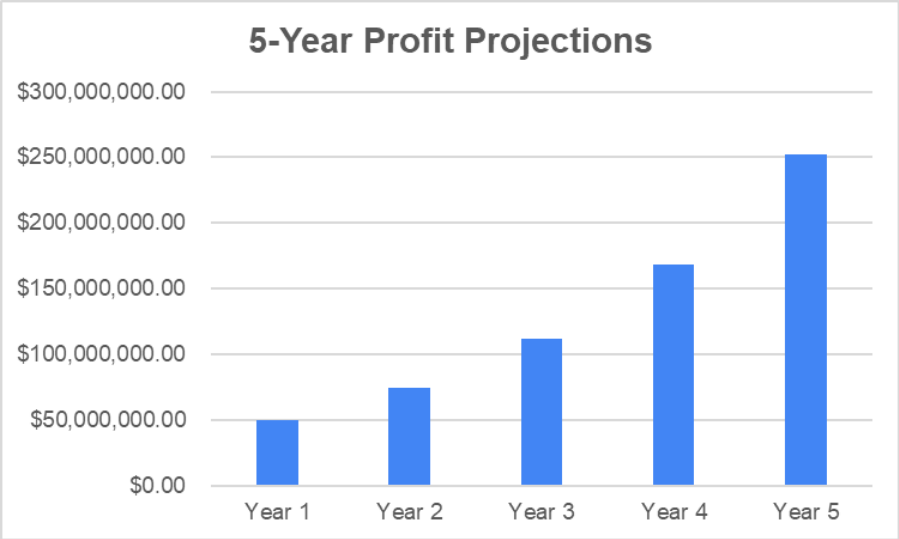

Figure 9

Competitor & Criteria	Affordability	Clean Energy	Small-Scale Flexibility	Efficiency
HyVera Tobe Energy	✓	✓	✓	~ 94%
H_2	✓	✗	✗	~ 75%
Bloomenergy nel	✗	✓	✓	~ 60%
SIEMENS energy	✗	✓	✗	~ 87%
Rockwell Automation	✗	✗	✗	~ 74%

Figure 10

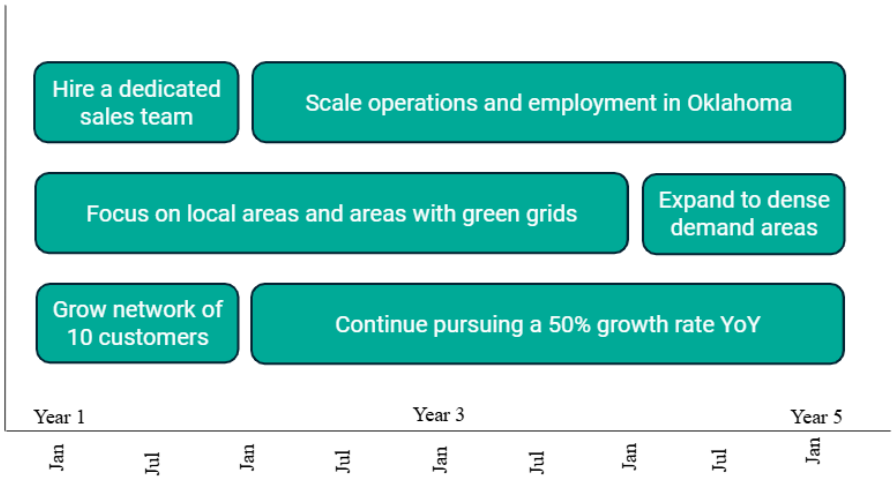

Table 1

Segment	Usage	Unit
1. **Heavy-duty fueling**: truck stops, long-haul corridors	4,000	kg/day
2. **Urban fleet fueling**: buses, municipal fleets, ports, airports	5,000	kg/day
3. **Light & medium-duty commercial fueling**: last-mile delivery fleets, local distribution	200	kg/day
4. **Industrial decarbonization**: electronics and semiconductor manufacturing, glass and metal processing, food and beverage hydrogenation	10,000	kg/day
5. **Chemical production & synthetic fuels**: ammonia, methanol production a. Small portion of the total industry that wants carbon offsets	15,000	kg/day
6. **Backup power & microgrids**: data centers, hospital backup, remote sites	100	kg/day

7. **Energy storage/grid balancing**: power-to-gas, seasonal/utility storage	1,500	kg/day
8. **Already green**: Tobe can undercut competitors and gain customers already committed to green hydrogen	3,100.00	kg/day

Table 2

Haas Revenue		
H2 Price (including compression)	3.51	per kg
25kg Installation fee	1,500.00	per device per contract
250kg Installation fee	15,000.00	per device per contract
2500kg Installation fee	150,000.00	per device per contract
Device Sales Revenue		
25kg Device price	43,125.00	per device
250kg Device price	600,000.00	per device
2500kg Device price	4,312,500.00	per device
25kg Device service	862.50	per d per year
250kg Device service	12,000.00	per d per year
2500kg Device service	86,250.00	per d per year

Table 3

HaaS Expenses		
H2 Cost	2.70	per kg
25kg Installation cost	1,500.00	per device
250kg Installation cost	15,000.00	per device
2500kg Installation cost	150,000.00	per device
Commision	0.18	per kg
Service cost	0.02	per device
Device Sales Expenses		
25kg Installation cost	1,500.00	per device
250kg Installation cost	15,000.00	per device
2500kg Installation cost	150,000.00	per device
25kg Commission cost	750.00	per device
250kg Commission cost	7,500.00	per device
2500kg Commission cost	75,000.00	per device
25kg Total cost	17,250.00	per device
250kg Total cost	240,000.00	per device
2500kg Total cost	1,725,000.00	per device

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